

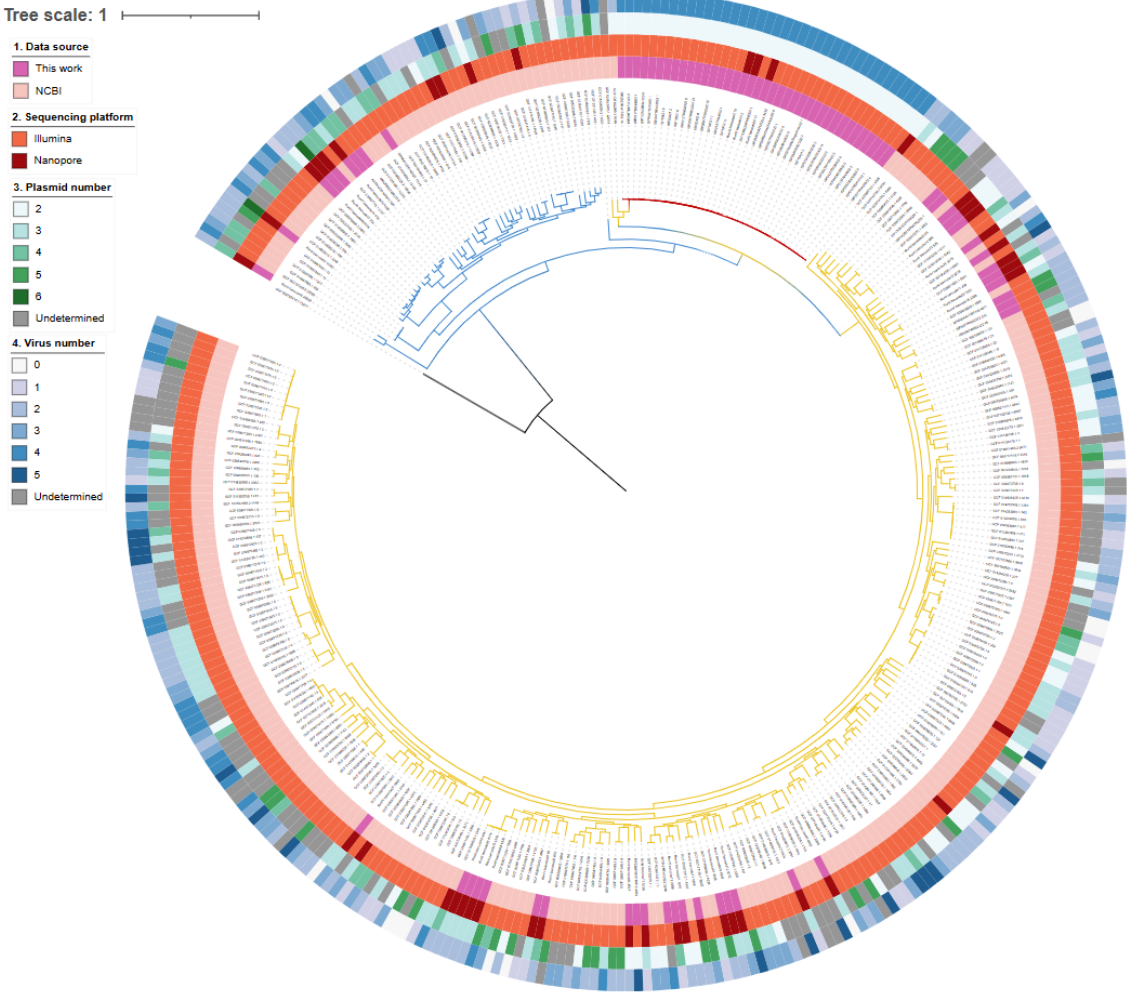


Fig. 1: Phylogeny and genome metadata of *Pantoea agglomerans* (PaggPan). Circular cladogram of the analyzed genomes. Concentric rings annotate (from inner to outer): data source (this study vs. NCBI), sequencing platform (Illumina vs. Nanopore), predicted plasmid count and predicted viral sequenced count (as indicated in the legend). Branch lengths are scaled to SNP distance (tree scale shown). Branch colours represent genomic clusters (C1 blue, C2 yellow and C3 red) determined by the PCoA for each genome (see Fig.2a).

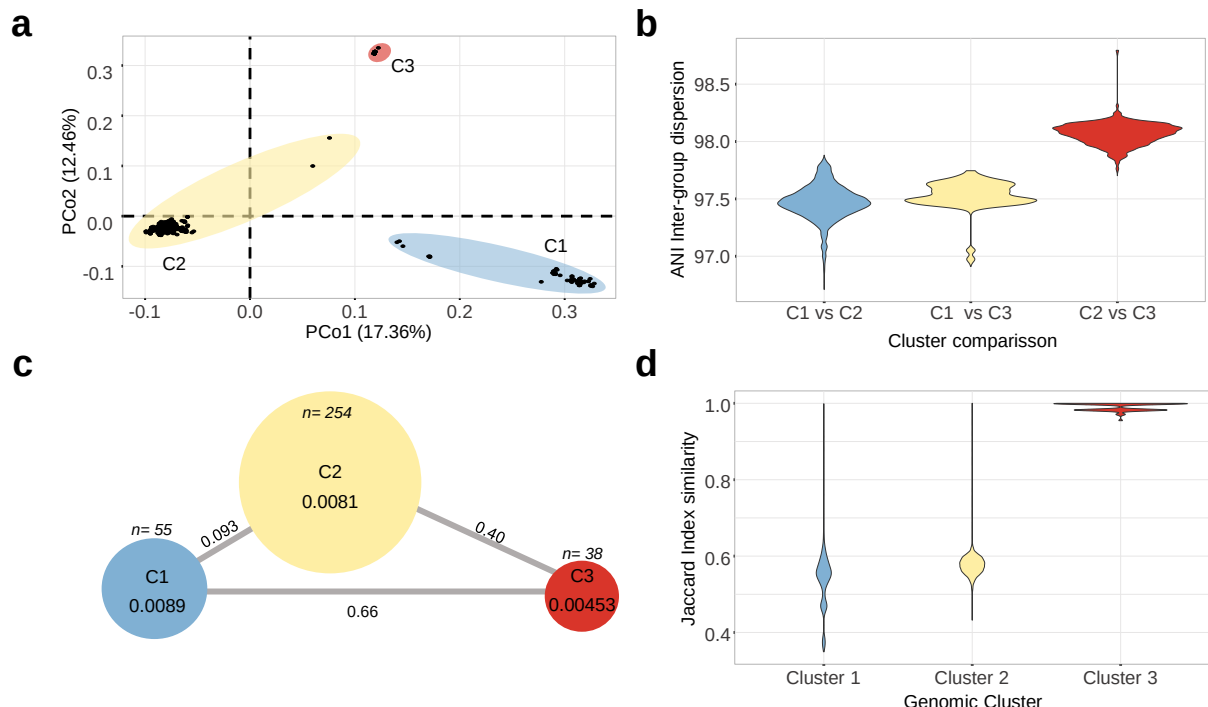


Fig. 2: PaggPan population structure resolves three divergent genomic lineages. **a**, Genome-wide average nucleotide identity (ANI) heatmap of 348 *P. agglomerans* genomes computed using BinDash, showing genomic clusters within the species. **b**, Principal coordinates analysis (PCoA) of the Jaccard index (JI) matrix of gene content similarity, revealing three major genomic clusters that together explain ~30% of the total variance. **c**, Violin plot showing the distribution of pairwise ANI values between genomes from different clusters. Inter-cluster comparisons are shown for C1–C2, C1–C3 and C2–C3. **d**, Mean nucleotide diversity (π) and pairwise F_{ST} values calculated between the clusters. Grey lines indicate F_{ST} values and the values within circles indicate mean nucleotide diversity (π) for each cluster. Circle size refers to the number of genomes per cluster (n). **e**, Distribution of pairwise Jaccard index (JI) values within each genomic cluster. Violin plots show that Cluster 3 displays higher JI values with reduced dispersion compared with Clusters 1 and 2.

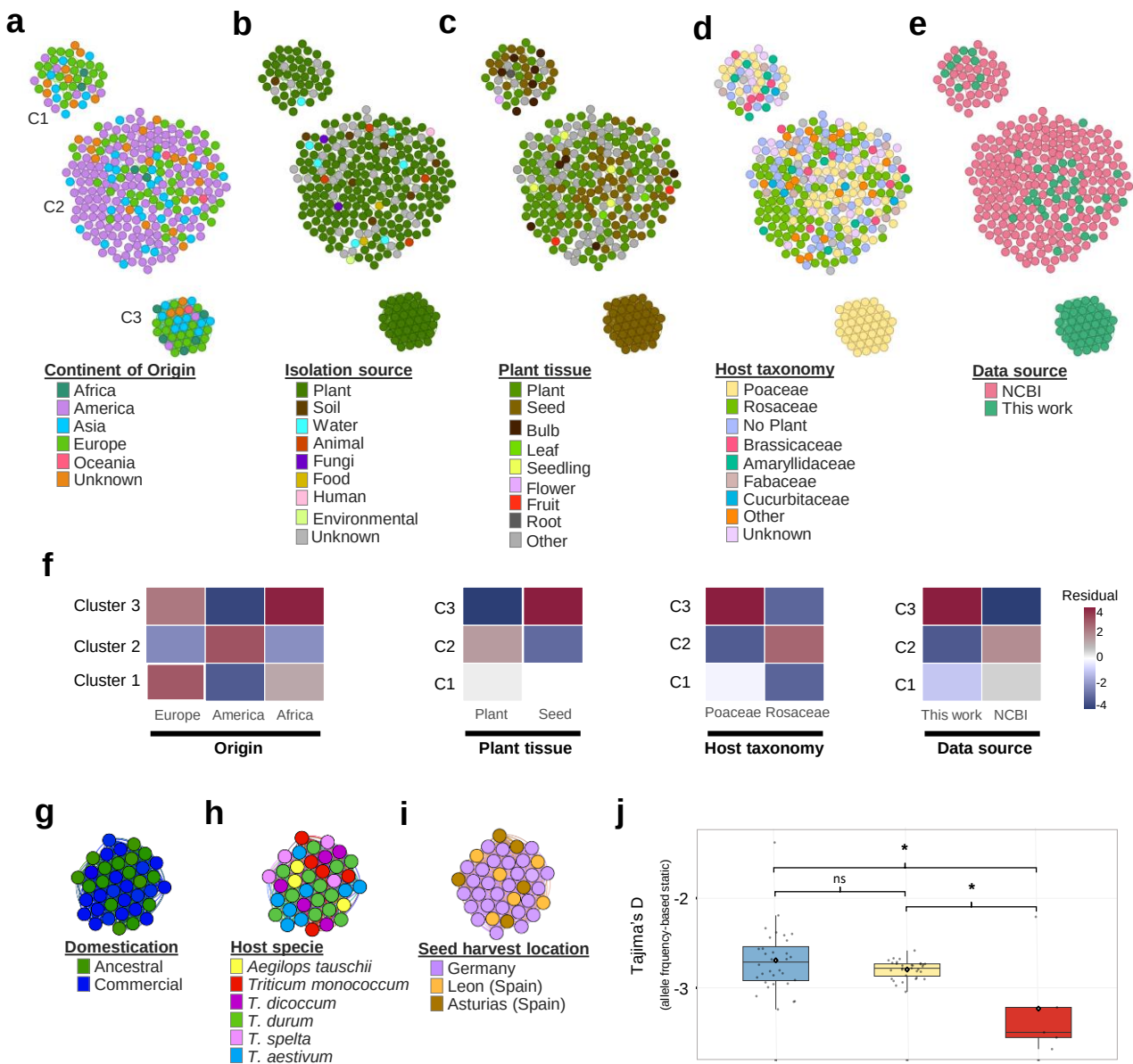


Fig. 3: Ecological and functional distribution of genomic lineages within the *Pantoea agglomerans*.

a-e Network-based representation of *P. agglomerans* genomes grouped according to genomic similarity. Each node represents a genome and node colours indicate different metadata categories or functional traits. Node position reflects network structure, and clusters (C1-C3) are indicated. The network is filtered at JI>0.7.

a, Geographic origin of isolates grouped by continent of original harvest location;

b, Isolation source metadata, including plant-associated, environmental, animal-associated and unknown sources;

c, Plant tissue of isolation,

d, Host plant family associated with each isolate.

e, Source of genomic data, distinguishing publicly available genomes retrieved from NCBI and genomes generated in this study.

f, Heatmap summarizing the relative enrichment and depletion of ecological metadata categories across genomic clusters. Colour scale represents standardized residuals, with red indicating enrichment and blue indicating depletion relative to expected frequencies, and all categories shown correspond to significant deviations (Benjamini–Hochberg adjusted $P < 0.05$).

g-i, High-resolution network representation of genomes belonging to Cluster 3 generated using a stringent Jaccard index threshold ($JI \geq 0.9$). Nodes represent individual genomes and are coloured according to wheat domestication status (**g**), wheat host species (**h**), and seed harvest location (**i**).

j, Box plots showing the distribution of Tajima's D measurements calculated in 10 kb windows for C1, C2 and C3. Boxes indicate interquartile ranges, center lines denote medians, and points represent individual windows per cluster (n indicated in the plot). Statistical significance was assessed using a Kruskal-Wallis tests followed by Dunn's post hoc tests with Bonferroni correction (ns, not significant; *, $P < 0.05$; ****, $P < 0.0001$).

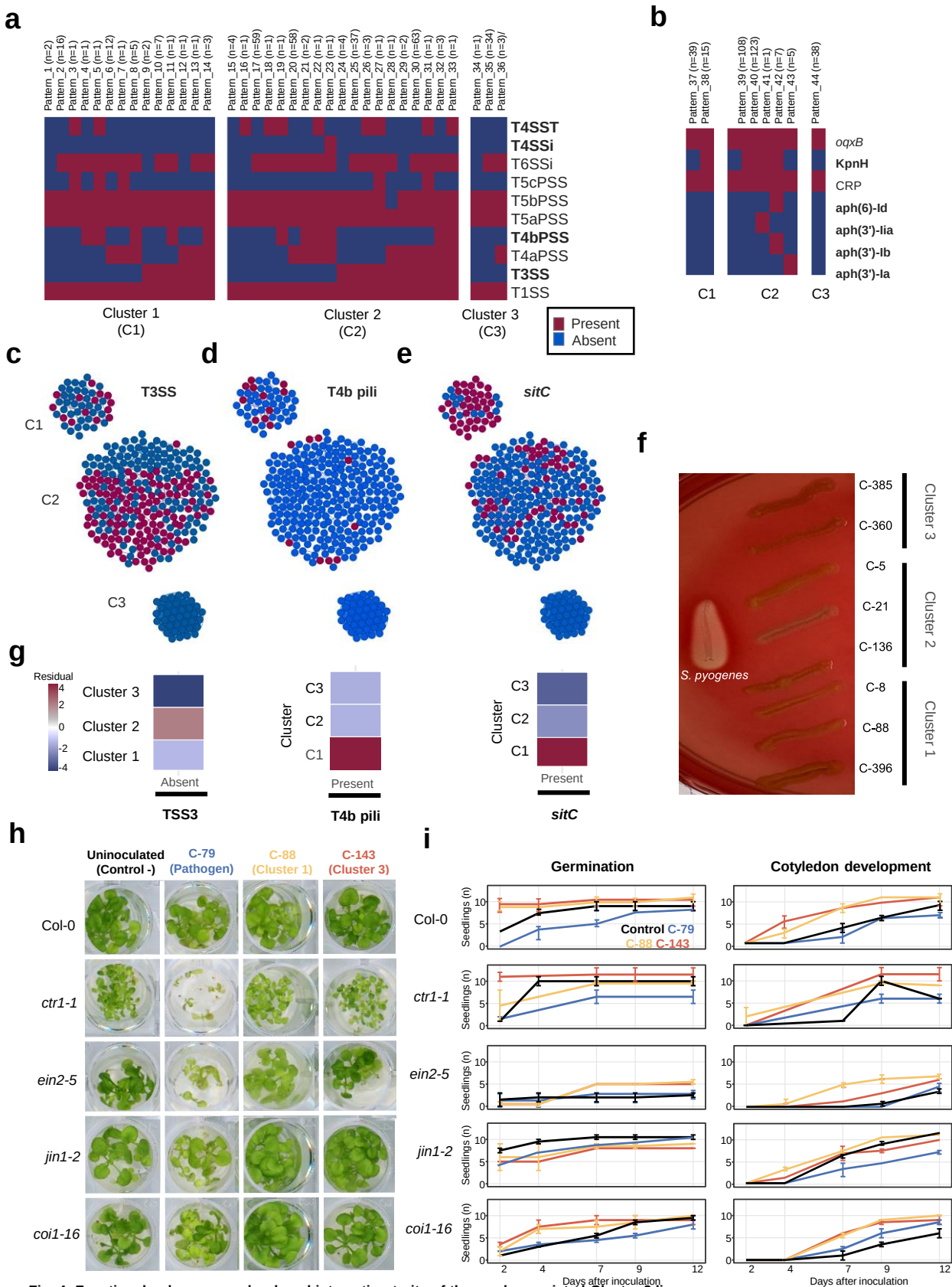


Fig. 4: Functional coherence and reduced interaction traits of the seed-associated Cluster 3 lineage.

a-b, Presence-absence heatmap of representative genome-encoded traits across *P. agglomerans* isolates. Columns represent associated secretion systems (a) or AMR genes (b), and rows represent genomic patterns (groups of genomes sharing identical trait presence-absence profiles). Identical genomic profiles were collapsed into representative patterns to simplify visualization. Patter identifiers are listed in Extended Data Table 5. Genomic features absent in C3 are shown in bold.

c-e, Presence/absence distribution of the type III secretion system (T3SS), type IVb or sitC-associated genes across the three genomic clusters. **f**, Hemolysis assay performed on representative *P. agglomerans* strains from Clusters 1, 2 and 3 using sheep blood agar with the positive control *Streptococcus pyogenes*. **g**, Heatmap summarizing the relative enrichment and depletion of virulence-associated traits across genomic clusters. Colour scale represents standardized residuals, with red indicating enrichment and blue indicating depletion relative to expected frequencies, and all categories shown correspond to significant deviations (Benjamini-Hochberg adjusted $P < 0.05$).

h, Representative images of *Arabidopsis thaliana* seedlings grown under different bacterial treatments. Genotypes include Col-0 (wild type) and mutants impaired in ethylene (ctr1-1, ein2-5) and jasmonate (jin1-2, coi1-16) signalling pathways. Seedlings were either left uninoculated (control -) or inoculated with seed-associated bacterial isolates C-79 (*Erwinia persicina*, pathogenic control), C-88 (cluster 1) and C-143 (Cluster 3).

i, Quantification of germination and cotyledon development over time. Data represent the number of seedlings per plate (mean $\pm$ s.e.m.). Colours correspond to treatments shown in panel **h**. Experiments were performed with two independent biological replicates ($n=10$).

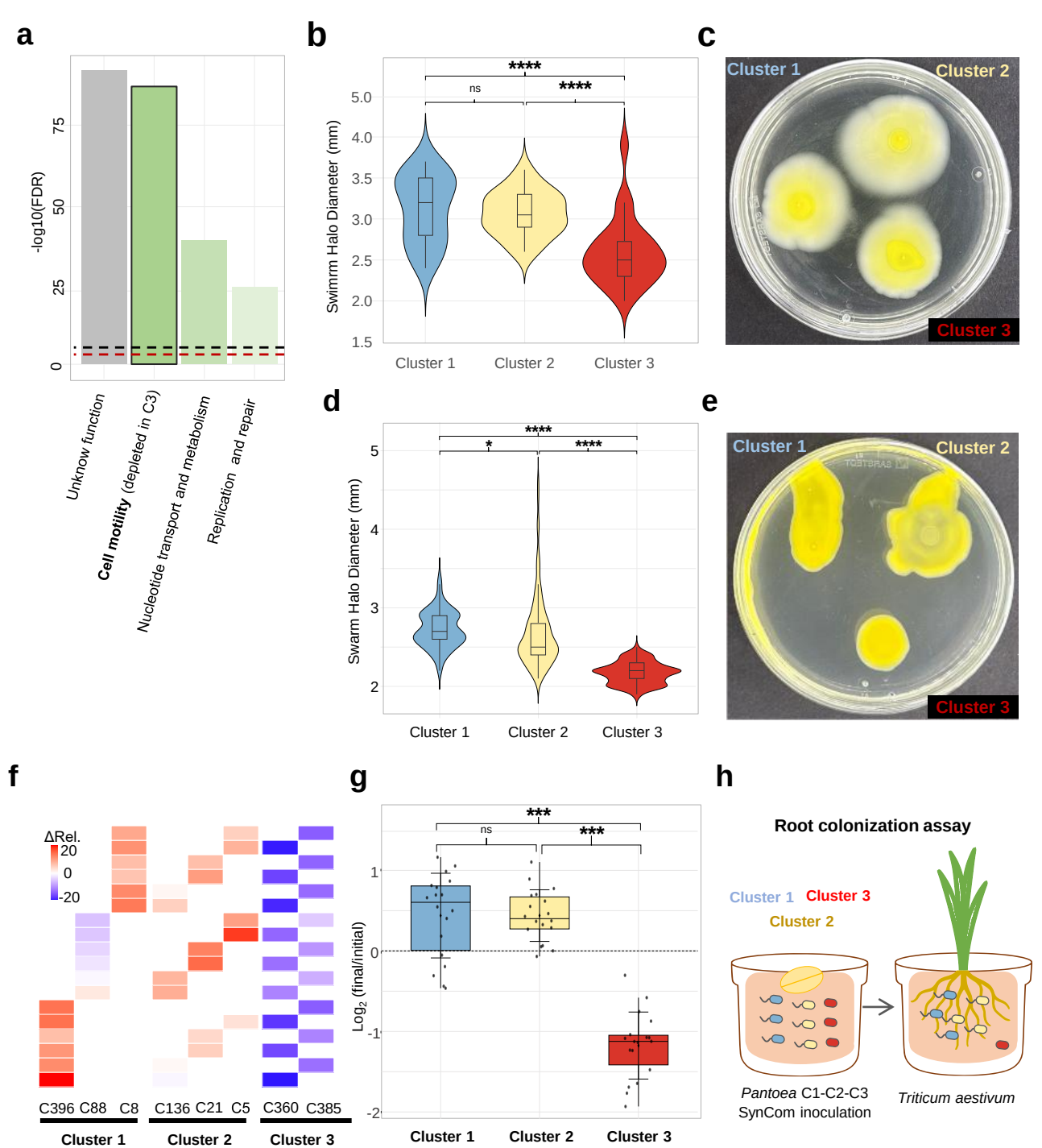


Fig. 5: Convergent gene loss underlies reduced motility and impaired root colonization in seed-specialist Cluster 3.

a, Top COG functional categories enriched among orthologous genes shared by Clusters 1 and 2 relative to Cluster 3 (Fisher's exact test with FDR correction). Bars show $-\log_{10}(\text{FDR})$; dashed horizontal lines denote FDR thresholds (black, $\text{FDR} < 0.05$; red, $\text{FDR} < 0.001$).

b,d, Quantification of motility phenotypes. Swimming (b) and swarming (d) halo diameters of isolates grouped by genomic cluster (C1, C2 and C3) were measured after 48 h (swimming, LB) and 72 h (swarming, glucose-supplemented LB). In b, data correspond to representative strains (C-88 (C1), C-136 (C2) and C-385 (C3)), whereas d summarizes all isolates shown in f. Violin plots show distributions; embedded boxplots (b) or overlaid points (d) indicate medians, interquartile ranges and biological replicates (b, $n = 25$; d, $n = 81$ for Clusters 1 and 2, $n = 48$ for Cluster 3). Pairwise Wilcoxon rank-sum tests (ns, not significant; *, $P < 0.01$; ****, $P < 0.0001$).

c,e, Representative plates corresponding to the assays in b and d. Images show C-88 (C1), C-136 (C2) and C3 (C-385 in c; C-360 in e).

f, Heat map showing changes in relative abundance ($\Delta\text{Rel.}$, difference between root-associated samples and the corresponding inoculum) of individual *Pantoea* strains across synthetic community (SynCom) combinations. Colours indicate relative enrichment (red) or depletion (blue) following root colonization (four biological replicates per SynCom).

g, Distribution of $\log_2$ fold changes in relative abundance between root-associated samples and inoculum for each genomic cluster. Boxes indicate interquartile ranges, centre lines denote medians, and points represent individual strain-SynCom observations across 18 biological SynComs ($n = 72$). Statistical significance was assessed using pairwise Wilcoxon rank-sum tests (ns, not significant; ***, $P < 1 \times 10^{-9}$).

h, Schematic summary of experimentally observed root colonization outcomes. Wheat seedlings were inoculated with SynComs comprising equal initial proportions of strains from Clusters 1, 2 and 3 and grown for 17 days in gnotobiotic FlowPots.

Seed transmission bottlenecks drive the emergence of a seed-specific endophytic lineage

